

Introduction to Differential Equations

1. State the order of the differential equations below.

(A) $x^3 y^5 + y' + 10x = 120$

Order =

(B) $10y' - x y'' + x^3 y^7 = 70$

Order =

(C) $(y')^5 + x y''' - x^5 \sin(y) = 2$

Order =

2. Show that the given function is NOT a solution to the differential equation.

(A) $x y' - 2y + 2x = 0$

$y = x^2$

(B) $y'' - 6y' + 9y = 0$

$y = e^{2x}$

(C) $y'' + 2y' + 2y = 0$

$y = e^{-x}$

3. Show that the given function is a solution to the differential equation.

(A) $x y' - 2y + 2x = 0$

$y = x^2 + 2x$

(B) $y'' - 6y' + 9y = 0$

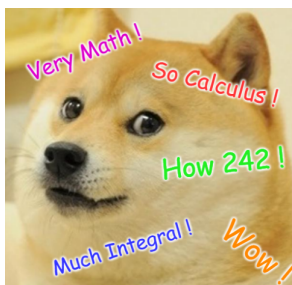
$y = x e^{3x}$

(C) $y'' + 2y' + 2y = 0$

$y = e^{-x} \cos(x)$



Scan the QR code!
Write an evaluation
for math 242!



4. Correctly label the slope fields:

(A) $y' = x$

(B) $y' = y$

(C) $y' = xy$

(D) $y' = -xy$

(E) $y' = \frac{y}{x}$

(F) $y' = -\frac{x}{y}$

(G) $y' = 1$

(H) $y' = x^2 - 1$

(I) $y' = y^2 - 1$

